

Azure Batch for an Engineering Simulation Workload

Problem statement

Engineering simulations often involve running a large number of calculations, and completing them one by one can take a lot of time and computing resources. In this project, the main problem is how to efficiently run around 300 simulation jobs without making the process too slow or expensive. Using a single system for all these jobs can create a heavy workload and increase the overall execution time.

Objective

The main objective of “Azure Batch for an Engineering Simulation Workload” is to develop a cloud-based system that can efficiently execute a large number of engineering simulation jobs in parallel using Microsoft Azure Batch.

- To execute approximately 300 simulation jobs using Azure Batch.
- To reduce execution time by distributing tasks across multiple compute nodes.
- To improve resource utilization through parallel processing.
- To provide a scalable solution that can increase or decrease computing resources based on workload.
- To reduce cloud computing costs by using low-priority/Spot compute nodes where suitable.
- To handle failed or interrupted tasks using automatic retry and rescheduling mechanisms.
- To manage simulation job dependencies and ensure tasks are executed in the correct order.
- To evaluate system performance based on execution time, successful jobs, failed jobs, retries, resource usage, and cost.

Abstract

The Azure Batch is a cloud-based High Performance Computing (HPC) platform which enables scaling of the computing processes. Engineering simulations typically require extensive computing resources due to the nature of the processes involved as several simulations need to be performed for different conditions. The project will utilize the Azure Batch to execute nearly 300 jobs simultaneously by tapping into the power of multiple virtual machines.

Parallel execution ensures that multiple simulation tasks can be accomplished simultaneously. This way, the total time required to complete the tasks is significantly reduced while also saving on computation costs since cheaper low-priority virtual machines can be leveraged to achieve the desired results. Additionally, the Azure Batch service allows easier management of the jobs as well as the underlying compute nodes. Virtual machines enable greater scalability, and therefore, can handle large volumes of simulation jobs. The approach is particularly useful in the context of engineering simulations as it typically involves several tasks which need to be performed concurrently. However, the parallel execution of tasks comes at some costs. Some of the challenges which were noted when using the Azure Batch for the project include node eviction. This phenomenon occurs especially with low-priority nodes when the Azure platform needs the resources which are being utilized by the simulations which have been deployed on the low-priority nodes.

Essentially, such nodes can be terminated at any time which may result in the loss of some of the computations. Another issue which may be encountered when utilizing Azure Batch for simulation tasks is job dependency modeling. This is particularly pertinent especially in the case of Azure Batch where the jobs need to be scheduled in relation to one another. In many cases, it may be necessary to perform simulations consecutively where the outcome of one simulation is used to determine the next steps. This becomes particularly problematic as the number of simulation tasks increases due to the heightened difficulty in scheduling the jobs as well as managing their dependencies. Overall, the project will demonstrate that Azure Batch can be leveraged as a cost-effective and

efficient platform for executing HPC tasks while also addressing the issues of job scheduling and dependency as well as the node eviction.

Use cases

- Engineering Simulation
- Scientific Computing
- Structural and Mechanical Analysis
- Automotive and Aerospace Testing
- Large-Scale Data Processing
- Cost-Optimized Cloud Computing

Requirements

- Microsoft Azure Account – To access Azure cloud services.
- Azure Batch – To create, schedule, and manage simulation jobs.
- Compute Pool – Multiple Azure compute nodes to execute tasks in parallel.
- Azure Storage – To store simulation input files and output results.
- Python 3.x – To create simulation tasks and automate job submission.
- Azure Portal – To configure and monitor the Azure resources.
- Internet Connection – Required to access and manage cloud resources

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